Deyuan (Mike) He

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Education

Princeton University, Princeton, NJ — 2022 – 2027 (expected)
Ph.D. in Computer Science · Advisor: Prof. Aarti Gupta
Research Interests: Formal Methods; Coding Agents; Distributed and Agentic Systems

University *of* Washington, Seattle, WA — 2018 – 2022
B.S. in Computer Science
Advisors: Prof. Zachary Tatlock & Dr. Steven Lyubomirsky

Selected Publications (*full list on Google Scholar*)

- ***crush*: Extensible Proof Automation for Lean**
  Mike He, D. Zhang, A. Gupta. *Under Review*, 2026.
- **Soteria: Formally Verified Planning with Runtime Enforcement for Safe LLM Agents**
  Mike He, A. Desai, S. Malik, A. Gupta. *NeurIPS*, 2026. (**Oral**)
- **Specy: Learning Specifications for Distributed Systems from Event Traces**
  Mike He, A. Desai, A. Jagarapu, D. Terry, S. Malik, A. Gupta. *OOPSLA*, 2026.
- **AlgoVeri: An Aligned Benchmark for Verified Code Generation on Classical Algorithms**
  H. Zhao*, Z. Yang*, J. Li*, **Mike He***, Z. Li*, C. Jin, V. Veeravalli, A. Gupta, S. Arora. *ICML*, 2026. (**Spotlight**)
- **Goedel-Code-Prover: Hierarchical Proof Search for Open State-of-the-Art Code Verification**
  Z. Li*, Z. Yang*, **Mike He**, H. Zhao, A. Zhao, S. Tang, K. Yang, A. Gupta, Z. Su, C. Jin. *COLM*, 2026.
- **Theory-Level Autoformalization: From Isolated Statements to Unified Formal Knowledge Bases**
  M. J. Min, **Mike He**, Z. Li, Z. Yi, S. Malik, A. Gupta, X. Si, O. Bastani. *ICML*, 2026. (**Spotlight**)
- **Verification of Compiler-to-Accelerator Mappings for Machine Learning Accelerators**
  A. Gaonkar, **Mike He**, Y. Li, B.Y. Huang, A. Cheung, V. Canumalla, G. Smith, et al.. *ArXiV*, 2026.
- **Dynamic Tensor Rematerialization**
  M. Kirisame*, S. Lyubomirsky*, A. Haan*, J. Brennan, **Mike He**, et al.. *ICLR*, 2021. (**Spotlight**)
- **Application-level Validation of Accelerator Designs Using a Formal SW/HW Interface**
  B.Y. Huang*, S. Lyubomirsky*, Y. Li, **Mike He**, G. H. Smith, T. Tambe, et al.. *TODAES*, 2024.
- **Ranking Formal Specifications using LLMs**
  Mike He, Z. Ang, A. Desai, A. Gupta. *LMPL Workshop*, 2025.

*: equal contribution.

Research Experience

Princeton University, Princeton, NJ — 2022 – Present
Graduate Research Assistant

- **Soteria** — Verified planning for LLM agents
  Built a Python-like DSL for composable plans, a verification-condition generator for temporal and relational data-flow properties, and runtime enforcement of verified plans.
  Result: **zero policy violations across 207 tasks** (τ^2-bench, Toolathon) with 4 LLM-backed agents.
- **AlgoVeri** — Cross-language verified code generation
  Curated aligned specifications for 77 classical algorithms in Dafny, Verus, and Lean 4.
  Result: best-model success varied from 40.3% (Dafny) to 24.7% (Verus) and 7.8% (Lean 4), identifying verification-language design as a dominant factor.

- **Goedel-Code-Prover** — LLM-guided proof search
  Curated the initial corpus from Tons of Inductive Problems; co-designed reward signals using proof context and Lean 4 verifier feedback.
  Result: an fine-tuned 8B model verified **62.0% of 427 tasks** (Verina, Clever, AlgoVeri), 2.6× the strongest baseline; outperformed frontier models (18.5%) and neural provers up to 84× larger.

Amazon Web Services, Santa Clara, CA & Boston, MA — Summer 2024, 2025, and 2026
Applied Scientist Intern, Agentic Automated Reasoning

- **Specy** — Specification learning for distributed systems (*2024 — 2025*)
  Led design and implementation in C# and Java: an event-quantified specification grammar and a learner combining enumerative search with dynamic learning from traces.
  Result: learned **all 43 target specifications** across 11 open-source and 3 AWS protocols (28 automatically, 15 with minor hints); recovered inductive invariants and exposed a production test-coverage gap. First-author paper at OOPSLA 2026.
- **PVerifier** — Deductive protocol verification (*2025*)
  Designed Proof Script, a proof-decomposition DSL for verification of distributed protocols.
  Result: proved safety and correctness properties, including consensus, for ring leader election, Two-Phase Commit, and an internal AWS service.
- **Strata** — Compositional concurrency verification in Lean 4 (*2026*)
  Built cooperative coroutine to model async-Python programs for compositional verification.

Taichi Graphics, Remote and Beijing, China — Jun 2022 – Sep 2022
Compiler R&D Intern

- **Taichi IR** — Compiler support for tensor types
  Introduced a standalone tensor type; adapted load/store forwarding, dead-code elimination, and reaching-definitions passes to tensor-typed expressions.

Intel, Strategic CAD Lab, Hillsboro, OR — Mar 2022 – Jun 2022
Formal Verification Research Intern

- **Pyrope** — Correct-by-construction hardware modeling
  Built a Python framework unifying hardware models and proofs, compiled to Dafny.
  Result: verified the multi-Montgomery reduction algorithm.

Technical Skills

- **Formal Methods:** Dafny, P, Z3 / SMT solving, Lean, specification inference, equality saturation
- **Agents:** Verified code generation, LLM-guided proof search, agent runtime enforcement
- **Programming:** Python, Java, C#, Rust · **Systems:** Distributed systems, compilers, program analysis

Teaching

- COS 516: Automated Reasoning about Software — TA, Princeton (Fall 2023 & 2024)
- CSE 505: Principles of Programming Languages — TA, University of Washington (Spring 2021)

Honors & Service

- SEAS Award for Excellence, 2026
- AI for Math Fund by Renaissance Philanthropy and XTX Markets, 2026
- CRA Outstanding Undergraduate Researcher Award, Honorable Mention (2022)
- Spotlight Papers: **ICML 2026** (×2), **ICLR 2021**
- Reviewer: NeurIPS, IEEE Transactions on Mobile Computing, AAE@KDD'25, SciPy'25
- Artifact Evaluation Committee: POPL'25, PLDI'24, POPL'24, MLSys'23, MICRO'21